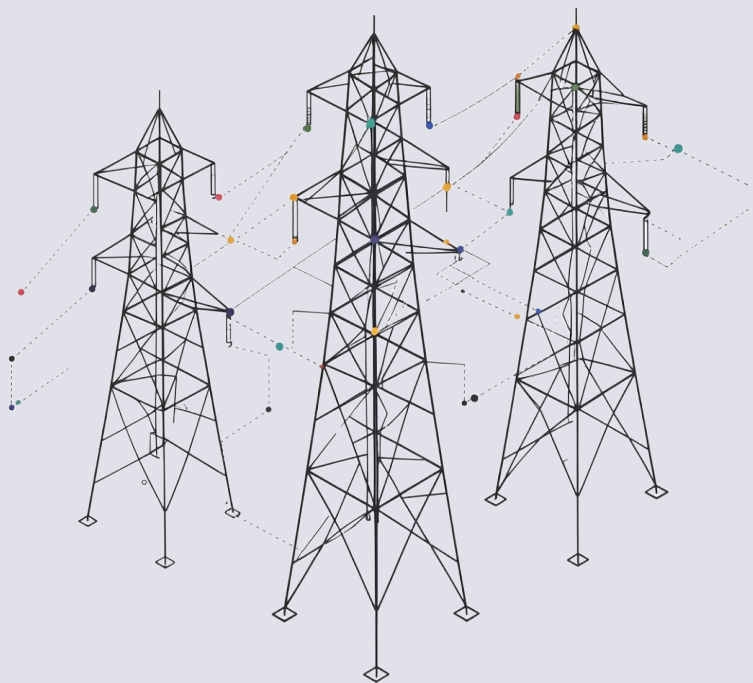
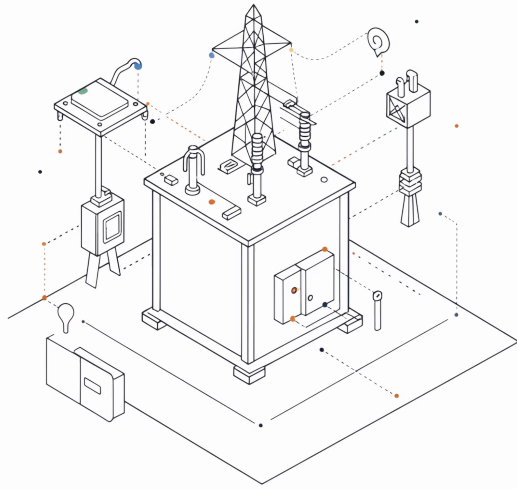




# GridGuard AI

*Predict. Prioritize. Prevent.*

**Track:** AI | **Team:** VertexAI | **Members:** Vaidehi Patel, Devanshi Patel, Aadya Raval, Durva Naik



# The Problem

Power-grid operators must monitor many assets, including transformers and substations. When equipment degrades, it can cause failures and outages. But knowing which asset may fail is not enough — operators also need to understand outage impact and what action to take.



**Core problem:** How can grid operators identify high-risk assets early, prioritize the assets that matter most, understand their potential impact, and take preventive action?

# The Solution

**GridGuard AI** is a grid-risk intelligence dashboard that analyzes equipment data and generates actionable insights across the full risk lifecycle.



Equipment Failure Risk  
Scores asset-condition risk



Outage Risk  
Evaluates operational outage risk



Risk Severity Labels  
Critical / High / Medium / Low classification



High-Risk Zones  
Identifies highest-risk assets and areas



Impact Analysis  
Quantifies consequences of potential failures



Operations Copilot  
Natural-language queries for operators



# Architecture

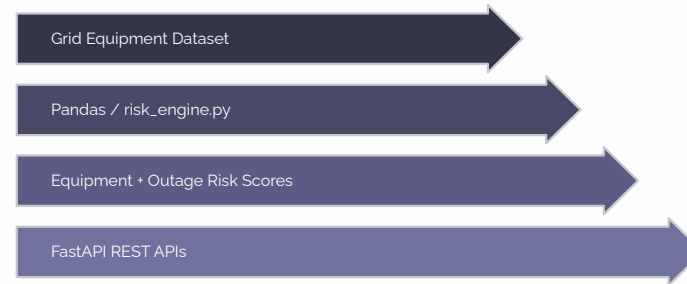
## Frontend

- Interactive GridGuard dashboard
- Risk visualizations
- Equipment details
- Impact & recommendations
- Copilot interface

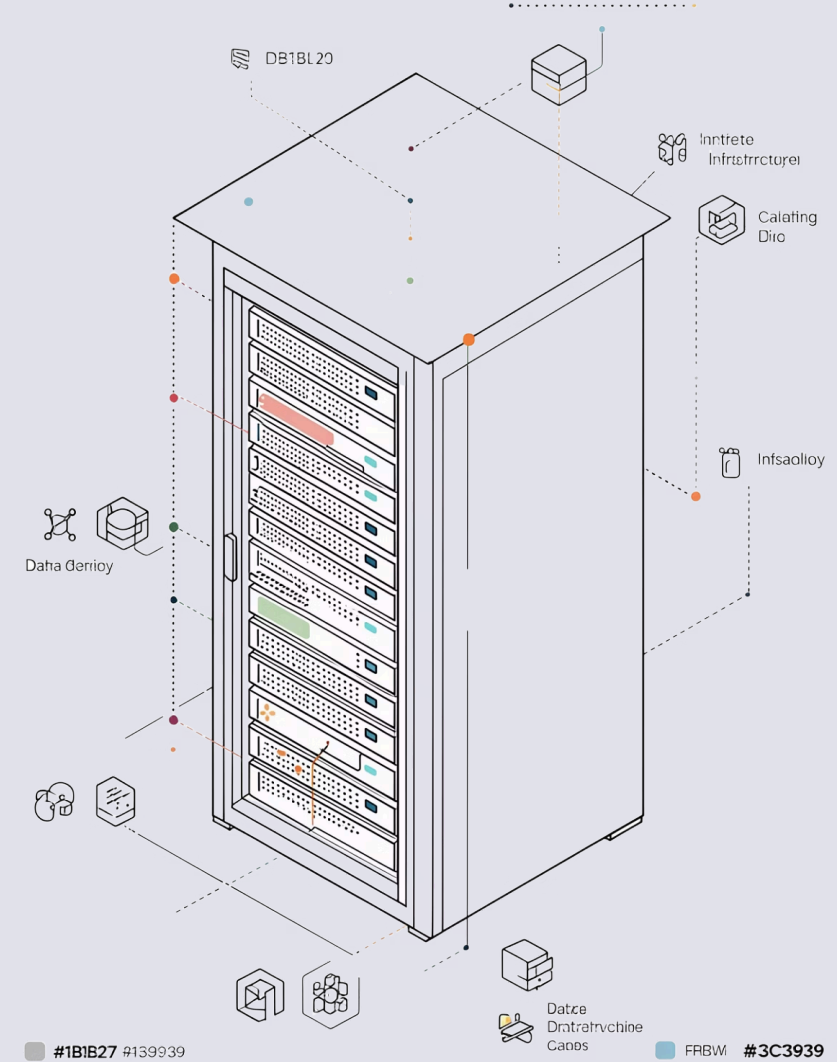
## Backend

- Python
- FastAPI
- Pandas
- In-memory risk processing

## Data Flow



 **Key technical point:** Dataset is loaded and scored at startup. No database and no external API dependency in the current MVP.



# Best Demo / Key Features

- 1 Dual Risk Scoring
  - Equipment risk + outage risk
- 2 Zone-Level Risk Prioritization
- 3 Impact + Recommendations
- 4 TF-006

# IBM Technologies

IBM Bob accelerated development and iteration of the GridGuard MVP.

## IBM Bob

- AI-powered development assistant used during development
- Helped generate and iterate on code
- Accelerated frontend and backend work

## Operations Copilot

- Rule-based MVP powered by live backend risk data
- Built for future IBM AI integration

# Results & Impact

Current MVP demonstrates:

50 grid assets analyzed

2 complementary risk scores  
(equipment + outage)

Zone + asset prioritization

Impact + preventive recommendations

Impact statement:

GridGuard AI transforms raw grid equipment data into prioritized, actionable risk intelligence — helping operators move from reactive outage response toward proactive asset management.

# Team

Vaidehi Patel

Backend, risk engine, IBM Bob development

Devanshi Patel

Frontend & dashboard

Aadya Raval

Data, impact analysis & recommendations

Durva Naik

Documentation, presentation & demo

